

AI-Augmented Preconditioning for Jacobian-Free Newton-Krylov Solvers

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1. Problem Formulation and JFNK Baseline

Nonlinear system from discretized Navier–Stokes equations:

$$F(u) = 0, \quad u \in \mathbb{R}^N.$$

Newton linearization at iterate $u^{(m)}$:

$$J(u^{(m)}) \delta u^{(m)} = -F(u^{(m)}),$$

where $J = \partial F / \partial u$ is not formed explicitly in JFNK.

Krylov solve with right preconditioning:

$$JP^{-1}y = -F(u), \quad \delta u = P^{-1}y.$$

Preconditioning objective:

$$JP^{-1} \approx I,$$

so that eigenvalues of JP^{-1} cluster near 1, reducing GMRES iterations and improving robustness of implicit CFD solves.

2. Hybrid Learned Operator Paradigm

Classical preconditioners use fixed analytic structure. The proposed operator introduces an adaptive learned correction in a low-rank subspace:

$$z = P^{-1}r = D^{-1}r + U\alpha.$$

Components:

$D^{-1} \in \mathbb{R}^{N \times N}$: fixed diagonal baseline preconditioner,

$U \in \mathbb{R}^{N \times k}$: low-rank basis (dominant correction directions),

$\alpha \in \mathbb{R}^k$: adaptive coefficients predicted by a neural network.

Interpretation: $D^{-1}r$ supplies guaranteed cheap damping; $U\alpha$ restores missing long-range couplings not captured by the diagonal model.

3. Low-Rank Subspace Generation U

Let $\{r_i\}_{i=t-m+1}^t \subset \mathbb{R}^N$ be recent Krylov residuals. Form residual snapshot matrix

$$R = [r_{t-m+1} \quad \cdots \quad r_t] \in \mathbb{R}^{N \times m}.$$

Compute truncated SVD:

$$R \approx \tilde{U}_k \Sigma_k V_k^\top, \quad \tilde{U}_k \in \mathbb{R}^{N \times k}.$$

Set

$$U := \tilde{U}_k,$$

with $k \ll N$, capturing dominant residual-energy modes.

Operational policy:

- ▶ Update U periodically (every few nonlinear or linear iterations), not every GMRES step.
- ▶ Reuse U between updates to control overhead and maintain stable inference features.

4. Neural Feature Space and Architecture

The network predicts α from dimensionally reduced, physics-aligned features:

$$\phi_t = \left[U^\top r_t, \|r_t\|_2, U^\top r_{t-1}, U^\top r_{t-2}, x_t \right].$$

Input semantics:

- ▶ $U^\top r_t$: current residual coordinates in dominant correction subspace.
- ▶ $\|r_t\|_2$: global magnitude scale for normalization and trust control.
- ▶ $U^\top r_{t-1}, U^\top r_{t-2}$: compressed temporal dynamics.
- ▶ x_t : coarse state descriptors (e.g., timestep, flow regime indicators).

Network output and correction map:

$$\alpha_t = \text{NN}(\phi_t), \quad z_{\text{corr}} = U\alpha_t.$$

Thus the learned action remains constrained to a physically interpretable low-dimensional manifold.

5. Initialization and Warm-Start Dynamics

Early training requires physics-consistent initialization to avoid destabilizing updates.

Initialization target:

$$\alpha \approx W(U^\top r), \quad W \approx I_k.$$

This enforces initial behavior close to projection-based correction:

$$U\alpha \approx U(U^\top r),$$

so the network first reproduces linear subspace projection before learning nonlinear refinements.

Warm-start strategy:

- ▶ Initialize each inference with localized historical features in ϕ_t .
- ▶ Use recent projected residual history to reduce transients after basis refresh.
- ▶ Preserve continuity across Newton steps by reusing hidden-state-like temporal context in feature space.

6. Offline/Online Training Methodology

Data acquisition: Run baseline JFNK/GMRES solves and record tuples

$$(r_t, U_t, x_t, z_t^*),$$

where z_t^* is an optimal correction proxy extracted from converged or high-quality GMRES behavior.

Target in reduced coordinates:

$$\alpha_t^* \approx \text{proj}_{U_t}(z_t^* - D^{-1}r_t),$$

with

$$\text{proj}_U(v) = U^\top v \quad (\text{for orthonormal } U).$$

Training objective:

$$\min_{\theta} \sum_t \|\text{NN}_{\theta}(\phi_t) - \alpha_t^*\|_2^2 + \lambda \mathcal{R}(\theta),$$

avoiding explicit Jacobian formation while teaching the model to emulate high-value correction directions.

7. Operational Stability and Asymptotic Complexity

Stability constraint:

$$\|\alpha\|_2 \leq C\|r\|_2,$$

implemented via output clipping, norm-aware scaling, or constrained final-layer parameterization.

Then

$$\|U\alpha\|_2 \leq \|U\|_2 C\|r\|_2,$$

which prevents unbounded corrective amplification and reduces divergence risk.

Complexity profile per application:

$$D^{-1}r : O(N), \quad U\alpha : O(Nk), \quad U^\top r : O(Nk).$$

Total dominant cost remains

$$O(Nk), \quad k \ll N,$$

with neural inference cost marginal relative to sparse operator evaluations and Krylov orthogonalization.

8. Synthesis and Deployment Envelope

$$z = D^{-1}r + U \cdot \text{NN}\left(U^{\top}r; \|r\|; \text{history}; x\right)$$

Deployment strengths:

- ▶ Steady and mildly unsteady flows.
- ▶ Implicit time stepping with recurrent linear-solve structure.
- ▶ Moderate Reynolds numbers where dominant correction subspaces evolve smoothly.

Failure modalities and caution regime:

- ▶ Highly turbulent, broadband dynamics with rapidly rotating dominant subspaces.
- ▶ Strongly intermittent transients causing feature distribution shift.
- ▶ Regimes requiring frequent U -updates that negate low-rank efficiency assumptions.

Practical rule: retain fallback to classical preconditioner if bounded-correction criteria or residual monotonicity checks are violated.